Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_solver_evolution.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.84516	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	77.5732	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1496.794	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.070818	0.070818	265.62958	0.549316	12.085

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.84516.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.

- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 77.5732.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1496.794.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.067409.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.070818.
- Held-out runtime ms: 265.62958.
- Robustness dispersion across held-out structure families: 0.004281.
- Worst held-out instances:
- X-n176-k26: feasible=True, penalized gap=0.098427, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.095804, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.057656, errors=[]

- X-n275-k28: feasible=True, penalized gap=0.056107, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.046096, errors=[]

Judge Note

Summary of Phase-9 CVRP Solver Evolution Suite

Solver Condition	Feasibility	Heldout Feasible Gap	Runtime (ms)	Robustness Dispersion	Notes
Baseline Nearest Neighbor Constructive	100%	0.2734	42.8	0.0028	Fastest runtime, largest gap
Baseline Clarke-Wright Savings	100%	0.0738	77.6	0.0078	Good gap and runtime
Baseline Regret Insertion + Local Search	100%	0.4664	1496.8	0.0869	Worst gap, much slower
Evolved Solver (phase9_solver_evolution)	**100%**	**0.0708**	265.6	0.0043	Best gap, moderate runtime, low robustness dispersion, high complexity and novelty

Conservative Interpretation

- **Feasibility:** All solvers, including the evolved one, maintain perfect feasibility (100%) on held-out families.
- **Objective Gap:** The evolved solver achieves the lowest feasible gap (0.0708), slightly outperforming Clarke-Wright Savings baseline (0.0738) and significantly better than others.
- **Runtime:** The evolved solver’s runtime (~266 ms) is higher than the two fastest baselines (42.8 ms and 77.6 ms) but much lower than regret insertion local search (~1497 ms).
- **Robustness:** Robustness dispersion of the evolved solver is quite low (0.0043), better than Clarke-Wright (0.0078) and regret insertion (0.0869), and close to nearest neighbor (0.0028).
- **Novelty and Complexity:** The evolved solver introduces substantial code novelty (0.55) and complexity (12.1), indicating more sophisticated heuristics.

Conclusion

The evolved phase-9 CVRP solver **does not clearly beat all fixed baselines across all axes**; it:

- Slightly improves the objective gap compared to the best baseline (Clarke-Wright Savings).
- Ensures robust feasibility and reasonable runtime (moderate, not the fastest).
- Exhibits higher code novelty and complexity.

It can be considered as a **competitive and robust solver with a slight advantage in solution quality over classical heuristics**, but runtime and complexity trade-offs should be weighed depending on application needs.